An Expository Paper on

Mathematical Bridges: Connections Between Complex Functions, Algebra, and Geometry

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Abstract

In this expository paper, we develop the structure of meromorphic functions on the Riemann sphere. We begin with an explanation of how the compactification of the complex plane by a single point at infinity leads to the extended complex plane, which is topologically a sphere as shown via stereographic projection. Using this framework, we show that every meromorphic function on the extended complex plane is a rational function, and we analyze how rational maps act on the sphere. In particular, we describe how a rational function of degree k determines a k-fold branched covering of the Riemann sphere. We then examine the special case $k = 1$, where the resulting bijections are precisely the Möbius transformations. Finally, we study the group structure of Möbius transformations and characterize their algebraic and geometric properties as automorphisms of the Riemann sphere.

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Section *1*

Introduction

The first use of complex numbers dates back to the mid 1500s when Gerolamo Cardano used them to find roots to cubic equations [6]. However it was not until the 1700s that the analysis of complex numbers began to take shape. In Euler's *Introductio in Analysin Infinitorum* (1748), Euler asserts that a variable can assume any value whatsoever, include an "imaginary one" [1]. Yet, it took until 1770 for Euler to give an explicit definition of "imaginary" numbers in *Vollständige Einleitung zur Algebra* [3].

Despite giving no explicit definition Euler worked with complex numbers in the factorization of polynomials. In section 30 of his *Introductio* he demonstrates that one can split a polynomial $p(x)$ by finding all of the roots to $p(x) = 0$, which include complex solutions [1]. Later in this same work Euler defines the complex exponential function e^z and its inverse $z = \log(y)$. It was through these definitions that he arrived at the Euler formulas

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin(z) = \frac{e^{iz} - e^{-iz}}{2}$$

The ideas presented in *Introductio in Analysin Infinitorum*, particularly those dealing with the definition of the complex logarithm, led to debate with Euler's correspondents, most notably D'Alembert. D'Alembert was not entirely convinced by Euler allowing negative values in the logarithm. Despite these disagreements D'Alembert used complex numbers with much success. In 1752 D'Alembert published his *Essai* which used complex functions to solve partial differential equations [1]. In this same work, we find the first instance of the Cauchy-Riemann conditions, where D'Alembert constructs a method for determining the velocity at any point of a fluid using

$$dq = Mdx + Ndz$$
$$dp = Ndx - Mdz.$$

Near the end of the 18th century complex analysis was being used with much success in the analysis of physical problems. In 1799 its use in pure mathematics reached new heights when Gauss published a proof of the Fundamental Theorem of Algebra in his doctoral dissertation. Even with the success of complex analysis in both pure and applied mathematics there was still hesitancy to admit complex numbers and complex functions as a real and central part of mathematics with Gauss himself saying that from the moment equations were found that do not admit roots other than those of the form $a + ib$, these "fictive quantities" have been introduced "into all of analysis." "By what right I will not discuss in this place" [1].

It was not until the advancements of Cauchy, who with others heightened mathematics to the level of rigor we still use today, that complex analysis became a rigorous mathematical tool. As early as 1814 Cauchy used the integration of complex functions over closed loops in the complex plane to yield powerful results and hinted at what would become the Cauchy integral formula. He would present his general integral formula in his 1827 *Résumé* paper which lay the groundwork for treating singular points of some function, using integrals, and demonstrating that the presence of singularities inside a contour effects the valuation of an integral.

It was alongside the developments of Cauchy that the geometric understanding of complex numbers began to take hold. Argand in 1806 introduced his Argand diagram, which viewed a complex number as a two dimensional vector on a plane, with some length in the real direction and some in the imaginary direction. In 1827 Möbius published his *Der barycentrische Calcul* which introduced transformations of the complex plane [9]. Möbius showed how these transformations preserve circles and lines in the plane, and mathematicians studying these transformations soon realized that one must adjoin the point at infinity for these transformations to be bijective.

In 1851 perhaps the single most significant advancement in complex function theory occurred when Bernhard Riemann published his doctoral dissertation. In his dissertation Riemann introduced the concept of what would later be called a Riemann surface, which is a two dimensional surface on which a multi-valued complex function becomes single valued. It is with Cauchy's advancements to the analysis of complex functions and Riemann's approach in introducing Riemann surfaces that this work truly begins, but we must start first with Cauchy and a review of the important results of complex analysis.

Slightly before this however, we must mention that the majority of this work is based upon Jones and Singerman unless otherwise noted [5]. Now we continue with our review.

1.1 A Review of Complex Analysis

We begin our review with key results of a first course in complex analysis. This section can be omitted when reading and referred back to as a reference as theorems are proven later in this paper. This section uses results from both [5][8]. We begin by recalling the definition of an analytic function.

Definition 1.1. A function f is *analytic at* z_0 if it is complex differentiable in a neighborhood of z_0 and if

$$f(z) = \sum_{j=0}^{\infty} \frac{f^{(j)}(z_0)}{j!}(z - z_0)^j$$

converges for all z sufficiently close to z_0.

We now recall the connection between a function analytic on an open set G and a power series uniformly convergent on G through the following definitions and theorems.

Definition 1.2. A series in z of the form $\sum_{j=0}^{\infty} a_j(z - z_0)^j$ with $a_j \in \mathbb{C}\ \forall j$ and $z_0 \in \mathbb{C}$ is called a power series. The constants a_j are the coefficients of the power series.

Theorem 1.3 (Radius of convergence)**.** *For any power series* $\displaystyle\sum_{j=0}^{\infty} a_j(z - z_0)^j$ *there is a real number* $R \in [0, \infty]$*, depending only on the coefficients* $\{a_j\}$ *and* z_0*, such that*

(i) the series converges for $|z - z_0| < R$*,*

(ii) the series converges uniformly on any closed subdisk $|z - z_0| \leq R' < R$*,*

(iii) the series diverges for $|z - z_0| > R$*.*

The number R *is called the* radius of convergence *of the power series.*

Theorem 1.4. *A power series sums to a function that is analytic at every point inside its circle of convergence.*

Theorem 1.5. *If $\sum_{j=0}^{\infty} a_j(z-z_0)^j$ converges to $f(z)$ in some circular neighborhood of z_0 (that is, the radius of its circle of convergence is nonzero), then*

$$a_j = \frac{f^{(j)}(z_0)}{j!} \qquad (j = 0, 1, 2, \dots).$$

Consequently, $\sum_{j=0}^{\infty} a_j(z-z_0)^j$ is the Taylor expansion of $f(z)$ around z_0.

And so we see that if a function is analytic in an open set G and its power series converges in G then $f(z)$ is equal to that power series in that open set. We can use the derivatives $f(z)$ to calculate the coefficients of the power series of f. If, instead of a disk we consider a function analytic in an annulus we instead have a Laurent series.

Theorem 1.6. *Let f be analytic in the annulus $r < |z - z_0| < R$. Then f can be expressed there as the sum of two series*

$$f(z) = \sum_{j=0}^{\infty} a_j(z-z_0)^j + \sum_{j=1}^{\infty} a_{-j}(z-z_0)^{-j},$$

both series converging in the annulus, and converging uniformly in any closed subannulus $r < \rho_1 \le |z - z_0| \le \rho_2 < R$. The coefficients a_j are given by

$$a_j = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{j+1}}\, d\zeta, \qquad (j = 0, \pm 1, \pm 2, \dots),$$

where C is any positively oriented simple closed contour lying in the annulus and containing z_0 in its interior.

The Laurent series of a function is important when classifying zeros and poles as often a function will be analytic in a neighborhood of a pole but not at the pole itself. We now review the results of the classification of zeros and poles

Definition 1.7. A point z_0 is called a *zero of order m* for the function f if f is analytic at z_0 and f and its first $m-1$ derivatives vanish at z_0, but $f^{(m)}(z_0) \neq 0$.

Theorem 1.8. *Let f be analytic at z_0. Then f has a zero of order m at z_0 if and only if f can be written as*

$$f(z) = (z - z_0)^m g(z),$$

where g is analytic at z_0 and $g(z_0) \neq 0$.

Definition 1.9. Let $G \subset \mathbb{C}$ be an open set and let $z_0 \in G$. A point z_0 is called an *isolated singularity* of a function f if f is analytic on a punctured disk

$$\{\, z \in \mathbb{C} : 0 < |z - z_0| < r \,\}$$

for some $r > 0$.

Definition 1.10. Let f have an isolated singularity at z_0, and let

$$f(z) = \sum_{j=-\infty}^{\infty} a_j (z - z_0)^j$$

be the Laurent expansion of f valid for $0 < |z - z_0| < R$. Then

(i) If $a_j = 0$ for all $j < 0$, we say that z_0 is a *removable singularity* of f;

(ii) If $a_{-m} \neq 0$ for some positive integer m but $a_j = 0$ for all $j < -m$, we say that z_0 is a *pole of order* m of f;

(iii) If $a_j \neq 0$ for an infinite number of negative values of j, we say that z_0 is an *essential singularity* of f.

Now that we have discussed analytic functions, and reviewed singularities, we recall what it means for a function to be meromorphic.

Definition 1.11. A function f is said to be *meromorphic* in a domain D if at every point of D it is either analytic or has a pole.

Theorem 1.12. *The meromorphic functions over $\mathbb{C}$ form a field with addition being the usual addition of functions and multiplication by a scalar in $\mathbb{C}$. Thus if f is meromorphic, then cf is meromorphic. If f and g are meromorphic then so are $f + g$ and $f - g$.*

Now that we have reviewed analyticity, zeroes, and singularities, we continue with the restatement of a few more useful theorems beginning with Cauchy's theorem.

Theorem 1.13 (Cauchy's Integral Theorem)**.** *Let f(z) be analytic in the simply connected region A and let γ be a piecewise differentiable closed path lying in A. Then*

$$\int_\gamma f(z)\, dz = 0.$$

As a consequence of Cauchy's theorem we get Cauchy's integral formula

Theorem 1.14 (Cauchy's Integral Formula)**.** *Let f(z) be analytic in a simply connected region A, let γ be a peicewise differentiable simple closed path in A and let b be a point of the region enclosed by γ. Then*

$$f(b) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{(z - b)}\, dz.$$

Furthermore, since an analytic function has a power series expansion which is absolutely convergent within its disk of convergence, we can differentiate under the integral sign to yield a generalized form of the integral formula

$$f^n(b) = \frac{n!}{2\pi i} \int_\gamma \frac{f(z)}{(z-b)^{n+1}}\, dz.$$

The next result gives us a useful tool to prove later theorems, by giving us a criteron by which analytic functions must be constant.

Theorem 1.15 (Liouville's Theorem)**.** *If f is analytic throughout $\mathbb{C}$ and if there is a real number M such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$, then f is constant.*

If an analytic function, instead of being bounded, accumulates zeros then the next result follows.

Theorem 1.16. *Let f be analytic in a region R with zeros at a sequence of points z_i which tend to a limit $z^* \in R$. Then f is identically zero in R.*

The next result deals with conditions when a complex function defines an open mapping. Since some complex functions have points with multiplicity, this next result also considers such functions.

Definition 1.17. A function $f : X \to Y$ between two topological spaces is called an *open mapping* if for every open set $U \subseteq X$, the image $f(U)$ is an open subset of Y.

Open mappings are useful as they guarantee that the image of a region remains a region. If a complex function is an open mapping then this guarantees that the structure of the mapped set remains.

Theorem 1.18. *(i) Let $R \subseteq \mathbb{C}$ be a non-empty, path-connected, open set and let f be a non-constant analytic function defined on R. Then $f : R \to \mathbb{C}$ defines an open mapping.*

(ii) If $z_0 \in R$ and if $f(z_0) = w_0$ with multiplicity m then there exists a neighborhood N of z_0 such that for all $w \in f(N) \setminus \{w_0\}$, the set $f^{-1}(w)$ contains m points in N.

1.2 The Riemann Sphere

We now return to Riemann and the prototypical example of a Riemann surface: the Riemann sphere. Consider the 2-sphere as a subset of $\mathbb{R}^3$ given by

$$S^2 = \{(x_1, x_2, x_3) | x_1^2 + x_2^2 + x_3^2 = 1\}.$$

We can identify the complex plane with the plane through $x_3 = 0$ and any $z \in \mathbb{C}$ given by $z = x + iy$ as the triple $(x, y, 0)$. We will construct a projection from S^2 to $\mathbb{C}$ as follows. Let $N = (0, 0, 1)$ be the topmost point on S^2 and consider the line through N intersecting S^2 at $Q = (x_1, x_2, x_3)$ and $\mathbb{C}$ at $P = (x, y, 0)$. This line gives us the *stereographic projection* of $S^2 \setminus \{N\}$ onto $\mathbb{C}$, $Q \mapsto P$, as shown in Figure 1.1. Let $\pi : S^2 \setminus N \to \mathbb{C}$ denote the stereographic projection. We will construct an explicit formula for π and then show that π is a homeomorphism.

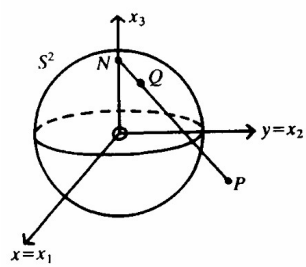

Figure 1.1: The Stereographic Projection[5].

Let O be the origin, then the points O,N,P form a right triangle ONP as seen in Figure 1.2. The point Q is on the hypotenuse of ONP, and if a line parallel to the x_2 axis is drawn connecting Q to the x_3 axis, then this triangle is similar to the triangle ONP. Therefore the sides are proportional and this yields the relationship

$$\frac{x}{x_1} = \frac{y}{x_2} = \frac{1}{1 - x_3}$$

Therefore the mapping of Q to P via π is given by $x = x_1/(1 - x_3)$, $y = x_2/(1 - x_3)$.

Theorem 1.19. *The mapping $\pi : S^2 \setminus N \to \mathbb{C}$ with $\pi : Q \mapsto P$ given by*

$$x = \frac{x_1}{1 - x_3}, \quad y = \frac{x_2}{1 - x_3}$$

is a homeomorphism.

Proof. First notice that π is continuous. Since we map $S^2 \setminus N$ then $x_3 \neq 1$ and x and y are each a ratio of continuous functions. Since ratios of continuous functions is continuous, the functions for x and y are continuous. Since $\mathbb{C}$ is topologically the product $\mathbb{R} \times \mathbb{R}$ we find that π is continuous. Now we proceed by showing that this function has continuous inverse π^{-1}.

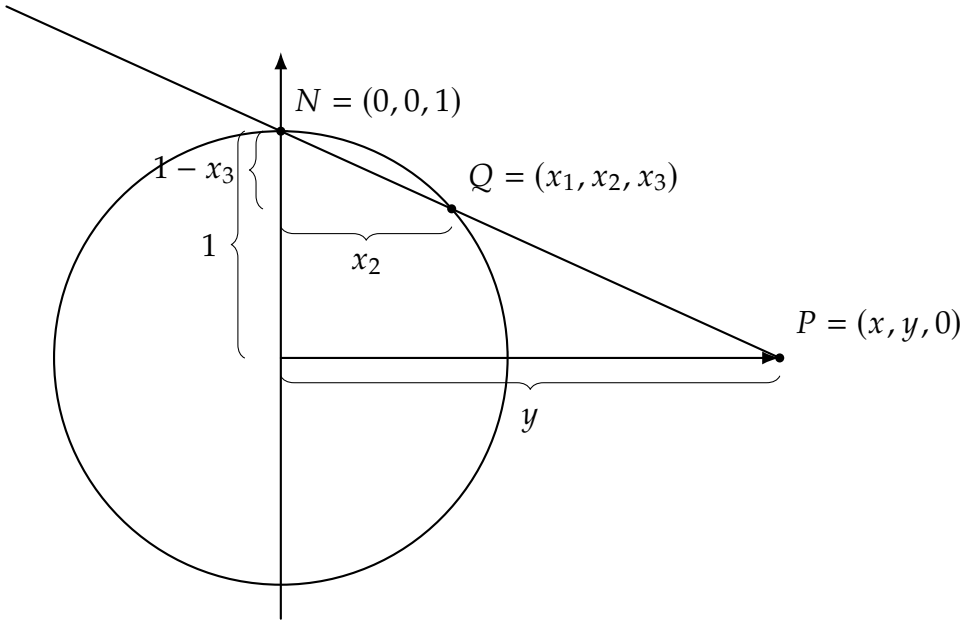

Figure 1.2: Stereographic projection viewed from the x_2-x_3 plane. Image generated using ChatGPT 5 from a hand drawn image of the author [7].

The condition for being a point on S^2 is $x_1^2 + x_2^2 + x_3^2 = 1$ we rewrite this as $x_1^2 + x_2^2 = 1 - x_3^2$. Now consider the formula for our projection given in the statement of the theorem and square the sum of x and y

$$x^2 + y^2 = \frac{x_1^2}{(1 - x_3)^2} + \frac{x_2^2}{(1 - x_3)^2} = \frac{x_1^2 + x_2^2}{(1 - x_3)^2}.$$

Substitute $x_1^2 + x_2^2 = 1 - x_3^2$ to yield

$$x^2 + y^2 = \frac{1 - x_3^2}{(1 - x_3)^2}.$$

We add unity to both sides and proceed through a sequence of algebraic steps to yield the equation

$$1 - x_3 = \frac{2}{x^2 + y^2 + 1}. \tag{$\star$}$$

Rearranging our formulas for x and y we have that $x_1 = x(1-x_3)$ and $x_2 = y(1-x_3)$. Substituting these formulas into ($\star$) yields

$$x_1 = \frac{2x}{x^2 + y^2 + 1}, \quad x_2 = \frac{2y}{x^2 + y^2 + 1}.$$

Furthermore we can rearrange ($\star$) and manipulate the expression to yield

$$x_3 = \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1}.$$

These three formulas are our inverse mapping π^{-1}. We show this by demonstrating that this function produces the identity in the first coordinate x_1. Showing that these formulas produce the identity in the other two coordinates proceeds exactly the same way. Since these formulas act as the identity in all three coordinates the formulas are the identity function as desired.

Consider $Q = (x_1, x_2, x_3) \mapsto P = (x, y, 0)$ with x and y given as in the statement of the theorem. Plugging these expressions into our formula for x_1 we have

$$\begin{aligned} x_1 &= \frac{2x_2/(1 - x_3)}{x_1^2/(1 - x^3)^2 + x_2^2/(1 - x_3)^2 + 1} \\ &= \frac{2x_2(1 - x_3)}{x_1^2 + x_2^2 + (1 - x_3^2)}. \end{aligned}$$

Applying $x_1^2 + x_2^2 + x_3^2 = 1$ in the denominator yields

$$\begin{aligned} \frac{2x_1(1 - x_3)}{2 - 2x_3} &= \frac{2x_1(1 - x_3)}{2(1 - x_3)} \\ &= x_1. \end{aligned}$$

Thus $\pi \circ \pi^{-1} = id_{S^2}$ and $\pi^{-1} \circ \pi = id_{\mathbb{C}}$ are continuous, with continuous inverses therefore π is a homeomorphism. □

We now wish to include N in our function in such a way that π remains a homeomorphism. Once we construct such a homeomorphism we retain the topological properties of S^2, while still being able to exploit the rich structure of $\mathbb{C}$. To do this we first introduce the extended complex plane.

Definition 1.20 (The Extended Complex Plane)**.** If $\mathbb{C}$ is the field of complex numbers then let $\Sigma = \mathbb{C} \cup \{\infty\}$ denote the union of $\mathbb{C}$ with the point at infinity. Then we call Σ the *extended complex plane*.

Corollary 1.21. *The mapping $\pi : S^2 \to \Sigma$ is a homeomorphism.*

Proof. Let $\pi : S^2 \setminus N \to \mathbb{C}$ be given as before and define $\pi(N) := \lim_{x_3 \to 1} \pi(Q) := \infty$. Then $\pi : S^2 \to \Sigma$ is a bijection. We use this to give a topology to $\mathbb{C} \cup \{\infty\}$. A set is open in $\mathbb{C} \cup \{\infty\}$ is open if and only if its pre-image under π is and subset $\mathbb{C}$ receives its standard topology. Since π is a bijection from two topological spaces, S^2 and $\mathbb{C} \cup \{\infty\}$, with continuous inverse, then π is a homeomorphism. □

We now refer to the mapping π as the stereographic projection from S^2 to the extended complex plane Σ. Since π is a homeomorphism, the topological properties of the 2-sphere are transferred to Σ. For this reason we hereafter refer to Σ as the Riemann sphere. We now show that Σ is compact.

Theorem 1.22. Σ *is compact.*

Proof. Recall that the Heine-Borel theorem states that a subset X of R^n is compact if and only if it is closed and bounded. As S^2 is closed and bounded it is therefore compact. Since $\pi : S^2 \to \Sigma$ is a homeomorphism, then Σ must be compact. $\square$

Section 2

Meromorphic Functions on the Riemann Sphere

Slightly before the work of Riemann appeared Laurent developed important ideas in complex analysis. Pierre Alphonse Laurent, was a contemporary of Cauchy, who, inspired by the work of Cauchy published an 1843 paper on the representability of holomorphic functions by a power series within an annulus (ref. Theorem 1.6). In doing so Laurent allowed the power series representation of holomorphic functions to have negative powers. This enabled the study of these functions near singular points and allowed for their classification [1].

Not long after, in 1850 Victor Puiseux would publish a paper in Liouville's *Journal* which introduced a distinction between branch points and poles of a function. It is with these poles that we are first concerned, now that we have allowed our space to include the point at infinity. We wish to include those functions with particular types of poles into our analysis of complex functions. Today we call these functions *meromorphic*. To return once again to Riemann's dissertation, he states:

> The unifying character of a class of functions which in the same way can be expressed through algebraic operations presents itself then in the form of the boundary and discontinuity conditions imposed on them. For example, if the range of variability of the quantity z extends simply or multiply over the entire infinite plane A, and within it the function is discontinuous only in isolated points, namely it permits only infinities whose order is finite, then the function is necessary algebraic [1].

And so it is towards this goal we begin by demonstrating how we can extend the definitions of functions to include the point at infinity. We do so in order that all points on Σ have equal status. Thus, if a function is analytic, meromorphic,

etc., on $\mathbb{C}$ we can then determine if that property is retained when we extend the functions definition to include infinity. To this end we define a new function $J : \Sigma \to \Sigma$ defined by $J(z) = z^{-1}$, with $J(0) := \infty$ and $J(\infty) := 0$ and study its properties.

First note that J is indeed a bijection, for it has an inverse, given by $J(z) = z^{-1}$ and since the function is self inverse it is an inversion. We want to determine how J acts on Σ and S^2 and so we return to our formulas for the stereographic projection $\pi : S^2 \to \Sigma$ given by $x = x_1/(1 - x_3)$, $y = x_2/(1 - x_3)$ with $z = x + iy \in \mathbb{C}$ and the formula for π^{-1} given by

$$x_1 = \frac{2x}{z\bar{z} + 1}, \quad x_2 = \frac{2y}{z\bar{z} + 1}, \quad x_3 = \frac{z\bar{z} - 1}{z\bar{z} + 1}$$

noting that $x^2 + y^2 = z\bar{z}$ since $z = x + iy$. Let $P \in \mathbb{C}$ be given by $z = x + iy$ then

$$J(P) = (x + iy)^{-1} = \frac{x - iy}{z\bar{z}}.$$

We let P^* denote $J(P)$ and we wish to examine the inverses of P and $J(P)$ respectively. Let $Q = \pi^{-1}(P)$ then Q is given by x_1, x_2, and x_3 above. Let $Q^* = \pi^{-1}(P^*) = \pi^{-1}(J(P))$ then

$$\begin{aligned}
x_1^* &= \frac{2x(z\bar{z})^{-1}}{(z\bar{z})^{-1} + 1} = \frac{2x}{1 + z\bar{z}} = x_1, \\
x_2^* &= \frac{-2y(z\bar{z})^{-1}}{(z\bar{z})^{-1} + 1} = \frac{-2y}{1 + z\bar{z}} = -x_2, \\
x_3^* &= \frac{(z\bar{z})^{-1} - 1}{(z\bar{z})^{-1} + 1} = \frac{1 - z\bar{z}}{1 + z\bar{z}} = -x_3.
\end{aligned}$$

Where we have used $x = \mathrm{Re}(z)$, $y = \mathrm{Im}(z)$ and thus $\mathrm{Re}(z^{-1}) = \mathrm{Re}(x - iy/z\bar{z}) = x/z\bar{z}$ and similarly for y, alongside

$$|J(P)|^2 = \left|\frac{x - iy}{z\bar{z}}\right|^2 = \frac{x^2 + y^2}{(z\bar{z})^2} = \frac{1}{z\bar{z}}.$$

We see that J maps z to z^{-1} on Σ, and J induces the transformation $\pi^{-1}J\pi : S^2 \to S^2$ which sends the point Q=(x_1, x_2, x_3) to Q* = $(x_1, -x_2, -x_3)$. This is a rotation of S^2 about the x-axis by π radians as seen in Figure 2.1.

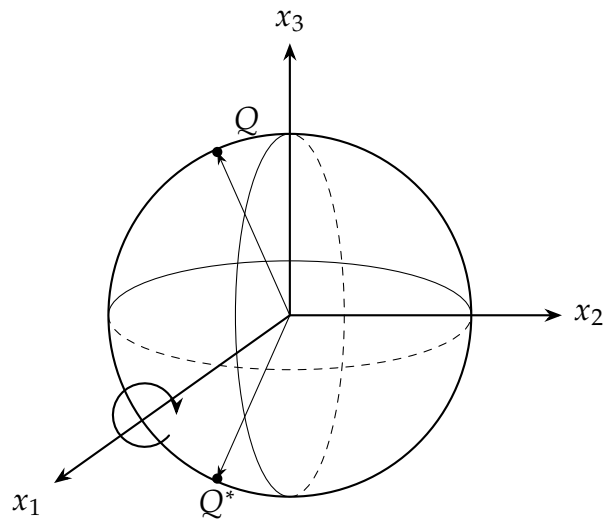

Figure 2.1: Point Q on the unit sphere and its image Q^* under a rotation by π about the x_1-axis. Figure generated using ChatGPT-5 using a hand-drawn figure by the Author.[7]

In the same way that we use π to identify S^2 with Σ we will abuse notation and consider J and $\pi^{-1}J\pi$ both under the label J. We can draw a diagram of this relation as seen in Figure 2.2. Now that we have introduced J and understood its affect on Σ we can use it to study the behavior of functions near ∞.

$$\begin{tikzcd} S^2 \arrow[r, "\pi^{-1}J\pi"] \arrow[d, "\pi"'] & S^2 \arrow[d, "\pi"] \\ \Sigma \arrow[r, "J"'] & \Sigma \end{tikzcd}$$

Figure 2.2: The relation between the spaces S^2, Σ through π, J, and $\pi^{-1}J\pi$

We can now include infinity in the domain of functions by defining $f(\infty) := \lim_{z\to\infty} f(z)$. We use J by composing J with f and say that f has the property at infinity that $f \circ J$ does at 0. In this way, we can write $f(\infty) := \lim_{z^{-1}\to 0} f(z^{-1}) = \lim_{z\to 0}(f \circ J)(z)$. Using this approach we can extend Theorem 1.16.

Theorem 2.1. *Let f be an analytic function on a non empty path connected open set R of Σ. If f has zeros at an infinite sequence of points z_n in R with a limit $z^* = \lim_{n\to\infty} z_n$ in R, then f is identically zero on R.*

Proof. If $z^* \neq \infty$ then there exists some M such that whenever $n \geq M$, $z_n \neq \infty$. Therefore we can omit finitely many terms and assume that z_n is a sequence contained entirely within $\mathbb{C}$. Now $R' = R \setminus \{\infty\}$ is still a path connected open set since we are removing only a single point. Thus f is analytic with zeros at an infinite sequence of points with a limit in R' and is therefore identically zero by Theorem 1.16. Now if $\infty \in R$ then f is analytic at ∞ since it is analytic on R and vanishes identically. Then we have $f(\infty) = 0$ as required by continuity of f.

Now suppose $z_n^* = \infty$ then there exists some M such that whenever $n \geq M$, $z_n \neq 0$. Since f is analytic on R it is analytic on $\tilde{R} \setminus \{0\}$ which contains z_n when $n \geq M$. Since f is analytic on $\tilde{R}$, $(f \circ J)$ is analytic on $R^* = \{z^{-1} | z \in \tilde{R}\}$ which contains the sequence z_n^{-1} whenever $n \geq M$. Now $(f \circ J)$ has zeros at the points z_n^{-1} and this sequence has limit $J(z^*) = 0$ in R^* and so $(f \circ J)$ is identically zero in R^* and thus f is identically zero in $\tilde{R}$. Now if $0 \notin R$ then $R = \tilde{R}$ and we are done. On the other hand if $0 \in R$ then $f(0) = 0$ as required by continuity and so f is still identically zero in R. □

Now that we have considered how to include the point at infinity in the domain of a function f, we consider now how to include infinity in the image of f. To do this we first recall how to express f near a point a when $f(a)$ is a solution $f(a) = c$, a zero $f(a) = 0$ and a pole $f(a) = \infty$. Recall that if f is meromorphic at a then f has a Laurent series expansion near a of the form

$$f(z) = \sum_{-N}^{\infty} b_j (z - a)^j.$$

Definition 2.2. In the particular case where f takes on a finite value then the coefficients, $b_j = f^{(j)}(a)/j!$ of the negative powers of the Laurent expansion are all zero. Then f takes the form

$$f(z) = c + \sum_{j=k}^{\infty} b_j (z - a)^j.$$

near $z = a$ with $f^{(k)}(a) \neq 0$ for some $k \geq 1$. We call the least such k the multiplicity of the solution of $f(z) = c$ at $c = a$. Similarly if $a = \infty$ then we say that $f(\infty) = c$ with multiplicity k if $(f \circ J)(0) = c$ with multiplicity k.

Definition 2.3. We also recall that if f is meromorphic with a pole at a of order k then we have a Laurent expansion of the form

$$f(z) = \sum_{-k}^{\infty} b_j (z - a)^j$$

near $z = a$. We call $\sum_{-k}^{-1} b_j(z-a)^j$ the principal part of f at a. If $a = \infty$ with multiplicity k then we consider $f \circ J(0)$ which has a zero of order k.

Corollary 2.4. *A non-constant meromorphic function $f : \Sigma \to \Sigma$ takes on any given value $c \in \Sigma$ only finitely many times, counting multiplicities.*

Proof. Let f be a non-constant meromorphic function. Let $z \in \Sigma$ and suppose $f(z) = c$. Suppose N_z is an open neighborhood of z. Since f is a non-constant meromorphic function we can find a neighborhood N_z such that $f(z) \neq c$ in $N_z \setminus \{z\}$. For if $c = \infty$ then c is a pole of f and thus a zero of $f \circ J$, and thus must be isolated by Theorem 2.1. On the other hand if $c \neq \infty$ then f does not take on c in $N_z \setminus \{z\}$ since z is a zero of $g(z) = f(z) - c$ a meromorphic function. Due to the compactness of Σ we can take a finite subcover of these open sets $N_{z_1}, N_{z_2}, \cdots, N_{z_k}$ so $f^{-1}(c) = \{z_1, z_2, \cdots, z_k\}$ is a finite set and the solution c has multiplicity k. □

Theorem 2.5. *Let f and g be meromorphic functions on Σ with poles at the same points in Σ, and with the same principal parts at these points. Then $f(z) = g(z) + c$ for some constant c.*

Proof. Since the meromorphic functions form a field (Theorem 1.12) $h = f - g$ is meromorphic. Since Σ is a compact set and h continuous due to its meromorphicity, $h(\Sigma)$ is compact. Now since f and g had poles at the same points with identical principal parts, h has no poles on Σ, that is $h(\Sigma) \subseteq \mathbb{C}$. Thus h is a bounded function on a compact set in $\mathbb{C}$ and must be constant by Liouville's Theorem. Thus $h = c$ and $c = f - g$ and so $f = g + c$ as desired. □

Theorem 2.6. *Let f and g be meromorphic functions on Σ with zeros and poles of the same orders at the same points of $\mathbb{C}$ Then $f(z) = cg(z)$ for some constant $c \neq 0$.*

Proof. Suppose that neither f or g are identically zero (then they would be constant). Since neither are identically zero the functions f/g and g/f are defined. At least one of these (but possibly both) is finite at infinity. Suppose it is f/g and let $h = f/g$. Thus h is an analytic function on Σ. We apply the same approach to show that $h = c$ as in the previous Theorem 2.5. Thus $c = f/g$ and $f = cg$ as desired. □

2.1 Rational Functions

We now wish to sharpen our understanding of meromorphic functions. A new classification will allow us to more easily determine some of the characteristics we have discussed in the previous section, and open the door to proofs of new characteristics. It turns out that the classification we are looking for is rationality.

Definition 2.7. A function $f(z)$, $z \in \mathbb{C}$ is rational if it can be expressed in the form $f(z) = \frac{P(z)}{Q(z)}$ where P, Q are co-prime polynomials (recall that two polynomials are co-prime in $\mathbb{C}$ if they share no roots).

We now show that a function is rational if and only if it is meromorphic.

Theorem 2.8. *A function $f : \Sigma \to \Sigma$ is rational if and only if it is meromorphic.*

Proof. Suppose f is rational and we decompose f as a product of linear factors. Then f is a differentiable function except at $z = \infty, \beta_j, (1 \leq j \leq s)$ and therefore f is analytic on $\mathbb{C} \setminus \{\beta_1, \cdots \beta_s\}$. At each β_j the function f has a pole of order n_j, while at ∞, f has a pole if $\deg(p) > \deg(q)$ of order $\deg(p) - \deg(q)$, or is otherwise analytic. Thus f is meromorphic.

On the other hand suppose that f is meromorphic. Then f has a finite number of poles and at each pole β we can express f in the form

$$f(z) = (z - \beta)^{-n} h(z)$$

where there is a pole of order n at β and $h(z)$ is analytic. If we do so for each pole we get

$$f(z) = (z - \beta_1)^{-n_1} \cdots (z - \beta_s)^{-n_s} g(z)$$

and thus $g(z) = (z - \beta_1)^{n_1} \cdots (z - \beta_s)^{n_s} f(z)$. Since g is analytic on $\mathbb{C}$ it has a Taylor series expansion

$$g(z) = a_0 + a_1 z + a_2 z^2 + \cdots$$

now since f is meromorphic at ∞ so is g and thus

$$(g \circ J)(z) = a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots$$

is meromorphic at 0. Thus $(g \circ J)$ has a pole of order k for some k. Thus g only has a finite number of positive powers and must be a polynomial. Hence

$$f(z) = g(z)(z - \beta_1)^{-n_1} \cdots (z - \beta_s)^{-n_s}$$

is a rational function. □

Now that we know meromorphicity is equivalent to rationality of functions we can determine properties of rational functions. These will give us key insights into how these functions behave on the Riemann sphere. First we define the degree of a rational function and demonstrate a key fact about a function's degree.

Definition 2.9. Let $f(z)$ be a rational function. Then

$$f(z) = c\frac{P(z)}{Q(z)} = \frac{(z-a_1)^{n_1}(z-a_2)^{n_2}\cdots(z-a_l)^{n_l}}{(z-b_1)^{m_1}(z-b_2)^{m_2}\cdots(z-b_k)^{m_k}}.$$

Then the degree of the rational function f is said to be the highest power appearing in the expansion by linear factors of P, Q.

Theorem 2.10. *If $f : \Sigma \to \Sigma$ is a rational function of degree $d > 0$, then f takes each value $c \in \Sigma$ exactly d times, counting multiplicities.*

Proof. Let $f = p/q$ where p and q are co-prime polynomials. First we consider $f(z) = \infty$. If $z \in \mathbb{C}$ then $f(z) = \infty$ if and only if $q(z) = 0$. By the fundamental theorem of algebra this happens exactly $\deg(q)$ times. Now if $\deg(p) \leq \deg(q)$ then these are the only poles of f. If $\deg(p) > \deg(q)$ then f has a pole of multiplicity $\deg(p) - \deg(q)$ at $z = \infty$ (since f behaves like $z^{\deg(p)-\deg(q)}$ as $z \to \infty$). In either case the number of solutions to $f(z) = \infty$ is $\max(\deg(p), \deg(q))$ which is the degree of f.

Now suppose that $c \neq \infty$. Since $\deg(f) > 0$, f is not identically equal to c. Recall that if we have a function h with poles at points $z \in \mathbb{C}$ then $1/h$ has zeros at those $z \in \mathbb{C}$. Similarly if a function h has solutions $h(z) = c$ at $z \in \mathbb{C}$ then $k(z) = h(z) - c$ has zeros there. We note these considerations and use them to write the function

$$g = \frac{1}{f-c} = \frac{1}{p/q - c} = \frac{q}{p - cq}.$$

Now g has $\deg(f)$ and poles where $f(z) = c$. By the approach we used when $f(z) = \infty$ we find that there are $\deg(g)$ of these counting multiplicities. Now q and $p - cq$ are co-prime since p and q are, so $\deg(g) = \max(\deg(p), \deg(p - cq) = \deg(f)$ as required.

□

We now define what it means for a point to be multiple.

Definition 2.11. Let f be meromorphic at $a \in \Sigma$ and let $f(a) = c$. Recall that a is a multiple point for f if the equation $f(z) = c$ has a multiple solution at $z = a$; if $c \neq \infty$ then this is equivalent to $f'(a) = 0$, while if $c = \infty$ then this is equivalent to f having a pole of at least order two at a. All other points are called simple points for f.

Corollary 2.12. *Let $f : \Sigma \to \Sigma$ be a rational function of degree $d > 0$. Then*

(i) f has only finitely many multiple points in Σ;

(ii) The inverse image of f, $|f^{-1}(c)| = d$ for all but finitely many points $c \in \Sigma$, and $1 \leq |f^{-1}(c)| < d$ for the remaining points in c. Note: $|\quad|$ is used to denote the cardinality.

Proof.

(i) Since f is a rational function with degree greater than zero its derivative f' is a nonzero rational function. Recall that a point $z \in \Sigma$ is a multiple point of f whenever $f'(z) = 0$. Since f' is rational it has a finite number of zeros and hence f has a finite number of multiple points.

(ii) By Theorem 2.10 if $c \in \Sigma$ then $f(z) = c$ has the solutions $z = a_1, \cdots, a_r$ with multiplicities $k_1, \cdots, k_r$ satisfying $k_1 + \cdots k_r = d$. Thus $|f^{-1}(c)| = r$ where $1 \leq |f^{-1}(c)| \leq d$ and $|f^{-1}(c)| = d$ precisely when each a_i has multiplicity exactly 1. Since f has only finitely many multiple points by (i) the result follows.

□

The result of Theorem 2.10 and its corollary provide us a new way to define the degree of a rational function. We can define $\deg(f)$ purely set theoretically as the maximum value attained by $|f^{-1}(c)|$ as c varies over Σ. Corollary 2.12 tells us that given some non-constant meromorphic function f, most points $a \in \Sigma$ are simple points. Now that we have characterized multiple points as being finite, we discuss the topology of the Riemann sphere near these points.

2.2 Topology

We have used Laurent's expansion to its full effect and examined the multiple (or equivalently branch) points of Puiseux. We continue our discussion with comments on the topology of the Riemann sphere, mapped to itself via some non-constant meromorphic function, near such points.

Let f be a non-constant meromorphic function. Then if $f(a) = c$ with multiplicity k and $a, c \in \mathbb{C}$, Theorem 1.18 (ii) shows that f is a k-to-one mapping near a. Thus, if we consider some open set $a \in U$ then there exists some open set V such that if $c' \in V \setminus \{c\}$ then $f(z) = c'$ has exactly k solutions all of which are simple (occuring with multiplicity 1). In the case where $a = \infty$ we use $(f \circ J)(a)$, and when $c = \infty$, we use $(J \circ f(a)$, and come to the same conclusion. And so this allows us to extend Theorem 1.18 (i), which shows that meromorphic functions

on $\mathbb{C}$ are open, to meromorphic functions on Σ. We now examine these points in detail beginning with a definition.

Definition 2.13. Let f be a non-constant meromorphic function. A point of multiplicity $k > 1$ is called a branch point of order $k - 1$.

And so if we have such a meromorphic function and $f(a) = c$ with multiplicity k, then let $w = f(z) - c$ we can write

$$w = f(z) - c = (z - a)^k g(z), \quad g(a) \neq 0$$

and $g(z)$ is analytic. Thus near a, the function w behaves like the map $z \mapsto z^k$. In order to determine the inverse of a given value we must solve $z^k = w$, where the number of solutions depends on the value of k. We have the solutions

$$z = |w|^{1/k} e^{2\pi i \frac{j}{k}}, \quad j = 0, 1, \cdots, k - 1$$

and hence the branch point a is of order $k - 1$. By Corollary 2.12 (i) the number of branch points are finite, denote them by B. Thus, for $c \in \Sigma \setminus f(B)$ the equation $f(z) = c$ has only simple solutions. The number of these depend on the degree of f, d, and these solutions take the form $a_1, \cdots, a_d \in f^{-1}(c)$. Each of the a_j has a neighborhood W_j which is mapped homeomorphically onto a neighborhood of c, say V_j. Though we omit some details here this is indeed a homeomorphism because f is an open mapping, and thus its inverse is a continuous function. Since f itself is continuous and has a continuous inverse, it is a homeomorphism.

Since f is a homeomorphism, we can shrink each W_j as to suppose they are disjoint. Let $V = \cap_{i=1}^{d} f(W_j)$. Then $f^{-1}(V)$ is the disjoint union of the $f^{-1}(V) \cap W_j = z_j$. then f takes each z_j homeomorphically to V. This shows that V is covered d times by the W_j by V. Thus $f : \Sigma \setminus f^{-1}(f(B)) \to \Sigma \setminus f(B)$ is a covering map.

Definition 2.14. A continuous map $f : X \to Y$ is a covering map if every $y \in Y$ has a neighborhood V such that $f^{-1}(V)$ consist of disjoint open sets mapped homeomorphically onto V by f, in which case we call X a covering space of Y. If we include the branch points then $f : \Sigma \to \Sigma$ is a branched covering.

Figure 2.3 shows a branched covering map $f : \Sigma \to \Sigma$ defined by $f : z \mapsto z^2$. Since z^2 has a zero of order 2 at zero we see that there is a branch point of order 1 at 0. Now, $J \circ f \circ J = z^2$ and this has a zero of order 2 at 0. Thus $f(\infty) = \infty$ with multiplicity 2. Therefore there is also a branch point of order 1 at infinity. These are the only zeros and poles of z^2 and therefore we have a branched covering of

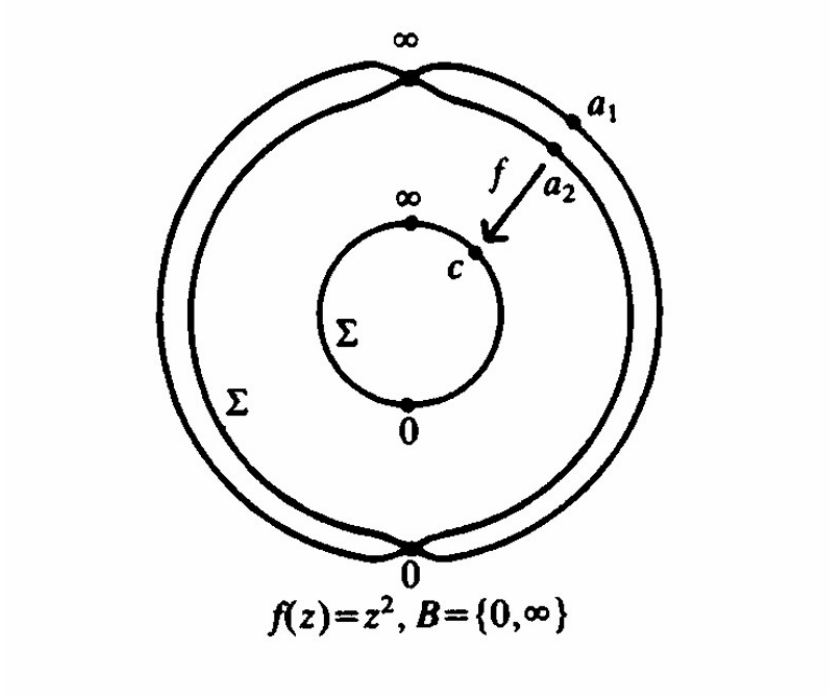

Figure 2.3: The branched covering $f : \Sigma \to \Sigma$ defined by $f : z \mapsto z^2$ with a zero of order 2 at zero and a pole of order 2 at ∞.

two sheets with two order 1 branch points.

We can begin to generalize this. In fact if we have $(z-c)^2$ then the previous analysis still holds, except that we now have a zero of order 2 at $z = c$ instead of $z = 0$. Likewise if we have $(z-c)^n$ our analysis still holds except we now have a degree n meromorphic function with zero of degree n and a pole of degree n at infinity, and our function represents a degree n covering. So let us now consider a rational (and therefore meromorphic) function of the form

$$f(z) = a\frac{(z-b)^n}{(z-c)^m}.$$

While we can clearly see that this function has a zero and pole of orders n and m respectively, and a pole at ∞ of order $n - m$ we proceed with the general approach. First note that if $n = m = 1$ then we have a bijection from Σ to Σ. If we assume that $m \neq n$ then the degree of the meromorphic function depends on

$\max\{n, m\}$. To determine the branch points we consider the derivative of f

$$\begin{aligned} f'(z) &= a\frac{n(z-b)^{n-1}(z-c)^m - m(z-b)^n(z-c)^{m-1}}{(z-c)^{2m}} \\ &= a\frac{(z-b)^{n-1}(z-c)^{m+1}(n(z-c)-m(z-b))}{(z-c)^{2m}} \\ &= a\frac{(z-b)^{n-1}(n(z-c)-m(z-b))}{(z-c)^{m+1}} \end{aligned}$$

Now since $b \neq c$ then if $z = b$, $z - c \neq 0$. So we have a critical point at $z = b$ of order n if $n \geq 2$ and the branch order is $(n-1)$. At $z = c$, f has a pole of order m if $m \geq 2$, and so this is a branch point of order $m - 1$. Now we see that f' has a third zero when

$$n(z-c) - m(z-b) = 0$$

or at

$$z^* = \frac{nc - mb}{n - m}$$

now to determine the order of this critical point we compute the second derivative to determine if $f''(z_0) = 0$. We have

$$\begin{aligned} f''(z) = a\,\frac{(z-b)^{n-2}}{(z-c)^{m+2}}\Big(&\big(n(z-c)-m(z-b)\big)\big((n-1)(z-c)-(m+1)(z-b)\big) \\ &+ (n-m)(z-b)(z-c)\Big). \end{aligned}$$

Now we plug in z^*

$$z^* = \frac{nc - mb}{n - m}. \qquad \text{So} \qquad n(z^*-c) - m(z^*-b) = 0,$$

the first term in parenthesis vanishes. We continue to simplify and find

$$\begin{aligned} f''(z^*) &= a\,\frac{(z^*-b)^{n-2}}{(z^*-c)^{m+2}}\Big(0 + (n-m)(z^*-b)(z^*-c)\Big) \\ &= a\,(n-m)\,\frac{(z^*-b)^{n-1}}{(z^*-c)^{m+1}}. \\ &= a\,(n-m)\,\frac{\left(\dfrac{n(c-b)}{n-m}\right)^{n-1}}{\left(\dfrac{m(c-b)}{n-m}\right)^{m+1}} \\ &\neq 0 \end{aligned}$$

since $c \neq b$ and $n \neq m$. And so we find that we have a Taylor series expansion of f near z^* of the form

$$f(z) = f''(z_*)(z - z^*)^2 + \cdots$$

where the first non-vanishing term is f'' and thus the order of the branch is $2 - 1 = 1$. We calculate the image, $f(z^*)$. To do this we first express the point z^* as follows

$$\begin{aligned} z^* - b &= \frac{nc - mb}{n - m} - b \\ &= \frac{nc - mb}{n - m} - \frac{b(n - m)}{n - m} \\ &= \frac{n(c - b)}{n - m} \end{aligned}$$

and similarly by subtracting c from z^* we get

$$z^* - c = \frac{m(c - b)}{n - m}.$$

Now plugging this into f yields

$$\begin{aligned} f(z^*) &= a\,\frac{(z^* - b)^n}{(z^* - c)^m}. \\ &= a\,\frac{\left(\frac{n(c-b)}{n-m}\right)^n}{\left(\frac{m(c-b)}{n-m}\right)^m} \\ &= a\left(\frac{n}{m}\right)^n \left(\frac{c - b}{n - m}\right)^{n-m}. \end{aligned}$$

and so z^* maps to $a\left(\frac{n}{m}\right)^n \left(\frac{c-b}{n-m}\right)^{n-m}$ under f. In particular, z^* is neither a pole nor a zero of f.

Now that we have considered general meromorphic functions, and have shown that functions are meromorphic if and only if they are rational, we will now look at a specific instance of a rational function, the Möbius transformation. In our example above is be the case where $m = n = 1$ and thus the function f is a bijection from Σ to Σ.

Section 3

Möbius Transformations

Alongside the developments of complex analysis discussed above, the mathematical community was undergoing another major change - it was becoming more German. While mathematicians in the earliest years of the 1800s would go to Paris for advanced studies, Germany quickly became the center of the mathematical world (at least according to Ohm) [4]. In 1838 Georg Simon Ohm (of Ohm's Law) drew up what he saw as a list of the most prominent contemporary mathematicians seen in Figure 3.1.

Germany		*France*	*Other nations*
Gauss	Möbius	Navier	Santini
Bessel	Schweins	Poisson	Plana
Jacobi	Schwerd	Cauchy	Ivory
M. Ohm	v. Staudt	Poinsot	Airy
Dirichlet	Desberger	Poncelet	Quetelet
Steiner	Littrow		
Plücker			

Figure 3.1: Ohm's 1838 List of Important Contemporary Mathematicians [4].

While we make no comment on the truth of Ohm's perspective (that is for historians of mathematics to decide) we use Ohm's list to center us in time. We can see in this list among Möbius the mathematicians who were contributing most to the development of complex analysis such as Gauss, Cauchy, Airy and so on. We also see that Riemann is not present. This was of course due to the fact that he was only 12 in 1838! And so it was during this time when Möbius was making his contributions; before the concept of a Riemann surface had been put

forth by Riemann.

As we have seen in the development of complex analysis thus far, many have viewed it as an algebraic discipline, while others saw it as geometric. If Möbius is to be put into one mathematical camp over another he would be a geometer, and it was through his geometric discoveries that he contributed to complex analysis - even if unintentionally. His geometric perspective was due to the fact that he was an astronomer as well as a mathematician. He wrote his doctoral thesis on *The Occultation of Fixed Stars* and published works on the principles of astronomy and celestial mechanics [9].

In his attempt to study barycentric coordinates (coordinates about the center of mass of an object) and projective geometry he introduced Möbius transformations. While he did not understand their connection to complex analysis and the Riemann sphere, it is our goal in this section to fully understand these transformations in the context of the previous sections - and in doing so reveal the connections to complex analysis and group theory hidden within. We begin with a definition and characterization of these transformations.

3.1 Characterization of Möbius Transformations

Definition 3.1. An automorphism of the Riemann sphere is a meromorphic bijection $T : \Sigma \to \Sigma$; we denote this set of functions by $\mathrm{Aut}(\Sigma)$

Now we need to determine what qualifies to be an automorphism of the Riemann sphere. As it turns out, the set of Möbius transformations qualifies.

Theorem 3.2. *$Aut(\Sigma)$ consists of the functions*

$$T(z) = \frac{az+b}{cz+d}, \quad (a, b, c, d \in \mathbb{C}, ad - bc \neq 0)$$

Proof. By Theorem 2.8, a function is meromorphic if and only if it is rational, and so T as defined above is in fact a meromorphic function. To see that it is bijective recall that a meromorphic function will cover the Riemann sphere according to its degree. Thus a meromorphic function is bijective on Σ if and only if it is of degree 1. Thus not both a and c can be zero. Now we only need to check the condition that $ad - bc \neq 0$. Recall that we require a rational function to be a ratio of two co-prime polynomials, thus $az + b$ and $cz + d$ must be co-prime. Equivalently they must share no roots, and solving for them

$$az + b = 0 \Rightarrow z = -b/a$$
$$cz + d = 0 \Rightarrow z = -d/c.$$

Therefore we require

$$\frac{b}{a} \neq \frac{d}{c}, \quad \text{equivalently}$$
$$ad - bc \neq 0.$$

Therefore the function is a ratio of two co-prime polynomials when the condition is satisfied. In the case where either $a = 0$ or $c = 0$ (recall that we cannot have both $a = 0$ and $c = 0$) we have either

$$f(z) = \frac{b}{cz + d},$$

or

$$f(z) = \frac{az + b}{d}$$

where $ad - bc$ is either $-bc \neq 0$ or $ad \neq 0$. And so again the condition is satisfied. □

Now that we have characterized the Möbius transformation as an automorphism of the Riemann sphere, we shall show that these transformations constitute a group.

Theorem 3.3. *The set of Möbius transformations given by*

$$T(z) = \frac{az + b}{cz + d}, \quad (a, b, c, d \in \mathbb{C}, ad - bc \neq 0)$$

form a group under function composition.

Proof. We first show that the composition of two Möbius transformations is again a Möbius transformation. Let U and T denote the transformations

$$U(z) = \frac{a'z + b'}{c'z + d'}, \quad T(z) = \frac{az + b}{cz + d}$$

Then we compute $(U \circ T)(z)$

$$\begin{aligned}(U \circ T)(z) &= \frac{a'\frac{az+b}{cz+d} + b'}{c'\frac{az+b}{cz+d} + d'} \\ &= \frac{\frac{a'az+a'b+b'cz+b'd}{cz+d}}{\frac{c'az+c'b+d'cz+d'd}{cz+d}} \\ &= \frac{(a'a + b'c)z + a'b + b'd}{(c'a + d'c)z + d'd + c'b}\end{aligned}$$

and so $U \circ T$ is a Möbius transformation if $(a'a + b'c)(d'd + c'b) - (a'b + b'd)(c'a + d'c) \neq 0$. We distribute these terms before factoring them into a more recognizable form

$$\begin{aligned} a'ad'd + a'ac'b + b'cd'd + b'cc'b - a'bc'a - a'bd'c - b'dc'a - b'dd'c \\ a'd'ad + a'c'ab + b'd'cd + b'c'cb - a'c'ba - a'd'bc - b'c'da - b'd'cd \\ a'd'ad - a'd'bc - b'c'ad + b'c'bc \\ (a'd' - b'c')(ad - bc) \neq 0 \end{aligned}$$

where the last line follows since each transformation guarantees one of the terms in the product is nonzero. Now we check for the existence of the identity transformation. We see if $a = d \neq 0$ and $b = c = 0$ we have $(ad - bc) \neq 0$ and

$$I(z) = \frac{az}{az} = z$$

as desired. Lastly, given $T(z)$ as above we find its inverse $T^{-1}(z)$. Let $T^{-1}(z) = \frac{dz-b}{-cz+a}$ then

$$\begin{aligned} T \circ T^{-1} &= \frac{a\frac{dz-b}{-cz+a} + b}{c\frac{dz-b}{-cz+a} + d} \\ &= \frac{\frac{adz-ab-bcz+ab}{-cz+a}}{\frac{cdz-bc-dcz+ad}{-cz+a}} \\ &= \frac{(ad-bc)z}{ad-bc} \\ &= z \end{aligned}$$

where the last line follows since $(ad - bc) \neq 0$. The reverse direction follows similarly and thus $T^{-1}(z)$ is a true inverse. Thus the set of Möbius transformations, $\mathrm{Aut}(\Sigma)$ is a group. □

Now that we know this fact we immediately get a corollary which strengthens our Möbius transformations as mappings.

Corollary 3.4. *$Aut(\Sigma)$ is a group of homeomorphisms from Σ to itself.*

Proof. Each Möbius transformation is a meromorphic function and hence continuous. Since $\mathrm{Aut}(\Sigma)$ is a group for each $T \in \mathrm{Aut}(\Sigma)$ we have $T^{-1} \in \mathrm{Aut}(\Sigma)$. Thus T^{-1} is also meromorphic and continuous. Therefore each T is a homeomorphism. □

Now that we understand that the Möbius transformations form a group of homeomorphisms, we begin a search to characterize this group more broadly. One

way that we can do this is to construct a homomorphism to another well-known group. Our homomorphism respects the group structure, and we can begin to gain insight about the group structure of Möbius transformations through our understanding of the group we map to.

To motivate this connection we consider a different representation of a given Möbius transformation. Let our transformation be given as before, by $T(z)$, then let M be the matrix given by placing the coefficients of our mobius transformation in the order that they appear, that is

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Now since we require of $T(z)$ that $(ad - bc) \neq 0$, we quickly see that in this representation this corresponds to $\det(M) = (ad - bc) \neq 0$. Therefore M is an element of the general linear group $GL(2, \mathbb{C})$. This suggests a mapping $\theta : \mathrm{Aut}(\Sigma) \to GL(2, \mathbb{C})$. We will show that this map is indeed a homomorphism, but first we recall the precise definition of the general linear group.

Definition 3.5. The general linear group of degree n is the set of $n \times n$ invertible matrices together with the operation of matrix multiplication.

Now we show that the mapping θ is a homomorphism of groups.

Theorem 3.6. *The mapping $\theta : GL(2, \mathbb{C}) \to Aut(\Sigma)$ defined by $\theta : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \frac{az+b}{cz+d}$ is a homomorphism of groups.*

Proof. Let T and U be the Möbius transformations

$$U(z) = \frac{a'z + b'}{c'z + d'}, \quad T(z) = \frac{az + b}{cz + d},$$

and let M, N be their matrix representations. Then

$$\begin{aligned} \theta(MN) &= UT \\ &= \frac{(a'a + b'c)z + a'b + b'd}{(c'a + d'c)z + d'd + c'b} \\ &= \frac{a'z + b'}{c'z + d'} \circ \frac{az + b}{cz + d} \\ &= \theta(M) \circ \theta(N). \end{aligned}$$

Thus θ is a group homomorphism. □

We note that in particular, θ is onto, since for any Möbius transformation, there is an element of $GL(2, \mathbb{C})$ which maps to it. Now that we have found a group homomorphism we would like to raise our homomorphism to the level of isomorphism, then we gain much more insight about the group structure of $\mathrm{Aut}(\Sigma)$ due to the structure preserving property of isomorphisms. To do this consider the kernel of θ, $\ker\theta$. This is the set of all elements of $GL(2, \mathbb{C})$ which act as the identity map in $\mathrm{Aut}(\Sigma)$.

The kernel of θ is the set of $GL(2, \mathbb{C})$ which map to $I(z) \in \mathrm{Aut}(\Sigma)$. Recall that the identity map is given by the conditions $a = d \neq 0$ and $b = c = 0$, and so $\ker\theta$ is the set of all matrices of the form

$$\begin{aligned}\ker\theta &= \left\{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a = d \neq 0, b = c = 0\right\} \\ &= \left\{\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \mid \lambda \in \mathbb{C} \setminus \{0\}\right\},\end{aligned}$$

where we have let $a = b = \lambda$. This is of course the set of all diagonal matrices, λI. If we take $GL(2, \mathbb{C})$ modulo $\ker\theta$ then we raise θ to the level of isomorphism,

$$\mathrm{Aut}(\Sigma) \cong GL(2, \mathbb{C})/\ker\theta.$$

This quotient $GL(2, \mathbb{C})/\{\lambda I | \lambda \in \mathbb{C}\}$ is exactly the definition of the Projective General Linear Group $PGL(2, \mathbb{C})$ where here 2 denotes dimension 2. And so we have

$$\mathrm{Aut}(\Sigma) \cong PGL(2, \mathbb{C}).$$

We now show an important property of Möbius transformations by first considering four general transformations of the extended complex plane. Since the function $f(\theta) = e^{i\theta}$, $\theta \in \mathbb{R}$ maps to the unit circle, if we simply fix some $\theta \in \mathbb{R}$ we get a fixed rotation along the arc of that circle of θ radians. We denote this map

(i) $R_\theta(z) = e^{i\theta} z$.

We have already seen that inversion $J(z) = 1/z$ is a rotation of Σ by π about the x axis, and this is our second transformation

(ii) $J(z) = 1/z$.

We can fix 0 and ∞ and scale by some arbitrary constant r such that $r \in \mathbb{R}$ and $r > 0$

(iii) $S_r(z) = rz$.

Finally we can fix ∞ and translate by some $t \in \mathbb{C}$

(iv) $T_t(z) = t + z$.

We will now see that any Möbius transformation can be written as a composition of these transformations. In demonstrating this we show that these transformations generate $\mathrm{Aut}(\Sigma)$.

Theorem 3.7. *Any Möbius transformation can be expressed as a composition of the transformations (i)-(iv).*

Proof. Recall that our Möbius transformation is of the form

$$T(z) = \frac{az + b}{cz + d}$$

where $ad - bc = 1$. Suppose $c = 0$ and then

$$T(z) = \frac{a}{d}z + \frac{b}{d}.$$

Let $a/d = re^{i\theta}$ and $b/d = t$ then

$$\begin{aligned} T(z) &= re^{i\theta} + t \\ &= T_t \circ S_r \circ R_\theta(z) \end{aligned}$$

as desired. On the other hand if $c \neq 0$ then we multiply and divide by c to get

$$T(z) = \frac{acz + bc}{c(cz + d)}.$$

adding and subtracting ad to the numerator allows us to apply the condition $ad - bc = 1$ as

$$\begin{aligned} T(z) &= \frac{acz + bc - ad + ad}{c(cz + d)} \\ &= \frac{a(cz + d) - (ad - bc)}{c(cz + d)} \\ &= \frac{a}{c} - \frac{1}{c(cz + d)}. \end{aligned}$$

Now, we let $t = a/c$ with $t \in \mathbb{C}$, then

$$T(z) = T_t \circ R_\theta \circ J(c^2 z + d).$$

Now $d \in \mathbb{C}$ and so let $t' = d$ then $c^2 \in \mathbb{R}$ with $c^2 > 0$ since $c \in \mathbb{C}$ by virtue that $t = a/c$, thus

$$T(z) = T_t \circ R_\theta \circ J \circ T_{t'} \circ S_r.$$

Therefore any Möbius transformation can be expressed in terms of the transformations (i)-(iv). □

3.2 Preservation of Circles and Lines

We will use the above fact to establish an important property of Möbius transformations. That is, that these transformations preserve circles and lines in the complex plane. We will first give a definition.

Definition 3.8. A circle, C, in the sphere $S^2 \subset \mathbb{R}^3$ is defined to be $S^2 \cap \Pi$ where Π is a plane in $\mathbb{R}^3$ and $|S^2 \cap \Pi| > 1$.

We see that a circle will always have more than one point in common with the plane that it intersects. This is due to the fact that we wish to exclude planes which are tangent to the sphere. A plane Π in $\mathbb{R}^3$ is given by the equation

$$\alpha x_1 + \beta x_2 + \gamma x_3 = \delta, \quad (\alpha, \beta, \gamma, \delta \in \mathbb{R})$$

we will apply π^{-1} the homeomorphism we defined in Section 1 which is given in coordinate form as

$$x_1 = \frac{2x}{|z|^2 + 1}, \quad x_2 = \frac{2y}{|z|^2 + 1}, \quad x_3 = \frac{|z|^2 - 1}{|z|^2 + 1}.$$

We wish to see where this plane is mapped to on the Riemann Sphere and so we plug in these expressions for x_1, x_2, and x_3 into the plane equation and get

$$\frac{2\alpha x}{|z|^2 + 1} + \frac{2\beta y}{|z|^2 + 1} + \gamma \frac{|z|^2 - 1}{|z|^2 + 1} = \delta.$$

We clear the denominator and bring everything to one side of the equation, proceeding to group like terms and simplify our expression into a more recognizable form. We obtain

$$2\alpha x + 2\beta y + \gamma |z|^2 - \gamma - \delta |z|^2 - \delta = 0, \quad \text{Thus}$$
$$2\alpha x + 2\beta y + |z|^2(\gamma - \delta) - (\gamma + \delta) = 0.$$

Let $a = \gamma - \delta \in \mathbb{R}$, $b = \alpha - i\beta \in \mathbb{C}$, and $c = -(\gamma + \delta)$ and use $x = \frac{z+\bar{z}}{2}$, $y = \frac{z-\bar{z}}{2i}$. We find

$$\alpha z + \alpha \bar{z} - i\beta z + i\beta \bar{z} + a|z|^2 + c = 0, \text{thus}$$
$$bz + \bar{b}\bar{z} + az\bar{z} + c = 0.$$

The final line above, $bz + \bar{b}\bar{z} + az\bar{z} + c = 0$, is the equation for a cline in the complex plane. A cline is either a circle or a line with the point at infinity, the latter being a circle in Σ. To see this more clearly we will return to the above derivation but expand in (x, y) coordinates instead of complex coordinates

$$\begin{aligned} 0 &= (\alpha - i\beta)(x + iy) + (\alpha + i\beta)(x - iy) + a(x + iy)(x - iy) \\ 0 &= ax^2 + ay^2 + 2\alpha x + 2\beta y + c. \end{aligned}$$

We will subtract c and group terms in x, y, then complete the square as follows

$$\begin{aligned} ax^2 + 2\alpha x + ay^2 + 2\beta y &= -c \\ x^2 + 2\alpha\frac{x}{a} + y^2 + 2\beta\frac{y}{a} &= -\frac{c}{a} \\ (x + \frac{\alpha}{a})^2 - \frac{\alpha^2}{a^2} + (y + \frac{\beta}{a})^2 - \frac{\beta^2}{a^2} &= -\frac{c}{a} \\ (x + \frac{\alpha}{a})^2 + (y + \frac{\beta}{a})^2 &= \frac{\alpha^2 + \beta^2 - ca}{a^2}. \end{aligned}$$

We see that this is in fact the equation for a circle with center at $(-\alpha/a, -\beta/a)$ and radius

$$\sqrt{\frac{a^2 + \beta^2 - ca}{a^2}}.$$

Now if $a \to 0$ or $a \to \infty$ then we get straight lines. Thus, circles in Σ are of two types: circles in $\mathbb{C}$ or sets of the form $\Lambda \cup \{\infty\}$ where Λ is a line in $\mathbb{C}$. We can connect $PGL(2, \mathbb{C})$ and circles in Σ with the following result. From now on when we refer to a circle in Σ we refer to both the circles in $\mathbb{C}$ and those circles in Σ which contain $\{\infty\}$.

Theorem 3.9. *If C is a circle in Σ, and if $T \in PGL(2, \mathbb{C})$, then $T(C)$ is a circle of Σ.*

Proof. Given some $T \in PGL(2, \mathbb{C})$, by Theorem 3.7, T can be represented as a composition of S_r, R_θ, T_t, and J. The generators J, R_θ are just rotations of Σ and therefore preserve circles. T_t, and S_r are translation and scaling, which both preserve circles. Therefore T as a composition of these elements preserves circles and lines. □

3.3 Transitivity and Cross-Ratio

Recall that a group G acts on a set Ω transitively, if for each $\alpha, \beta \in \Omega$, there exists some $g \in G$ such that $g(\alpha) = \beta$. More generally we can say that G acts $k-$transitively on Ω if whenever $(\alpha_1, \cdots, \alpha_k)$ and $(\beta_1, \cdots, \beta_k)$ are $k-$tuples of

distinct elements of Ω (that is, $\alpha_i \neq \alpha_j, \beta_i \neq \beta_j$ whenever $i \neq j$), there exists some $g \in G$ such that $g(\alpha_j) = \beta_i$ for $i = 1, 2, \cdots, k$. We see that transitivity is equivalent to 1-transitivity. We will show that $PGL(2, \mathbb{C})$ acts 3-transitively but not 4-transitively on Σ. We begin with the following theorem:

Theorem 3.10. *If z_1, z_2, z_3 are distinct elements of Σ, then there is a unique $T \in PGL(2, \mathbb{C})$ such that $T(z_1) = 0, T(z_2) = 1$ and $T(z_3) = \infty$.*

Proof. Let T be expressed as

$$\begin{aligned} T(z) &= \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} \\ &= \frac{(z_2 - z_3)z - z_1(z_2 - z_3)}{(z_2 - z_1)z + z_3(z_1 - z_2)} \end{aligned}$$

where $z_1, z_2, z_3 \neq \infty$. We can see that this is a valid Möbius transformation since

$$\begin{aligned} ad - bc &= z_3(z_2 - z_3)(z_1 - z_2) - z_1(z_2 - z_3)(z_1 - z_2) \\ &= (z_3 - z_1)[(z_2 - z_3)(z_1 - z_2)] \end{aligned}$$

and this is never zero, since z_1, z_2 and z_3 are distinct by hypothesis. If one of the $z_j = \infty$ then we simply take $\lim_{z_j \to \infty} T(z)$. Thus if $z_1 = \infty$ we have

$$\begin{aligned} \lim_{z_1 \to \infty} T(z) &= \lim_{z_1 \to \infty} \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} \\ &= \lim_{z_1 \to \infty} \frac{(\frac{z}{z_1} - 1)(z_2 - z_3)}{(1 - \frac{z_2}{z_1})(z_3 - z)} \\ &= -\frac{(z_2 - z_3)}{(z_3 - z)}. \end{aligned}$$

If $z_2 = \infty$ we have

$$T(z) = -\frac{(z - z_1)}{(z_3 - z)}.$$

If $z_3 = \infty$ we have

$$T(z) = -\frac{(z - z_1)}{(z_1 - z_2)}.$$

In each of these cases we see that we get a valid $T \in PGL(2, \mathbb{C})$. This choice of T does in fact send z_1 to 0 since if $z = z_1$ we get that the factor $(z - z_1)$ equals zero

and thus $T(z_1) = 0$, except in the case where $z_1 = \infty$ then we have $(z_3 - \infty)$ in the denominator and thus $T(z_1) = 0$. Now to see that $T(z_2) = 1$ we have

$$\begin{aligned} T(z_2) &= \frac{(z_2 - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} \\ &= \frac{-(z_1 - z_2)(z_2 - z_3)}{-(z_1 - z_2)(z_3 - z_2)} \\ &= 1. \end{aligned}$$

Lastly $T(z_3) = \infty$ due to the factor $(z_3 - z)$ in the denominator.

Now that we have shown such a T exists we show that it is unique. Suppose there exists some U with the same properties i.e $U(z_1) = 0, U(z_2) = 1, U(z_3) = \infty$. Then we can consider UT^{-1} which has the property that each of $0, 1$ and ∞ remain fixed. We will show that this is the identity. We have

$$UT^{-1}(z) = \frac{az + b}{cz + d}.$$

Now plug in 0 and we have that

$$UT^{-1}(0) = \frac{b}{d} = 0$$

implying that $b = 0$. Now plug in ∞

$$UT^{-1}(\infty) = \frac{a}{c} = \infty$$

and thus $c = 0$ (recall that $c = 0$ implies $a \neq 0$). Now consider $UT^{-1}(1)$

$$UT^{-1}(1) = \frac{a + b}{c + d}$$

now plugging in $b = 0, c = 0$ to find

$$UT^{-1}(1) = \frac{a}{d} = 1,$$

or $a = d$. Thus the Möbius transformation of $UT^{-1} = az/a = z$ which is the identity. Thus

$$UT^{-1} = I \Rightarrow U = T.$$

Therefore T is unique. $\square$

Now we can easily obtain two corollaries.

Corollary 3.11. *If (z_1, z_2, z_3) and (w_1, w_2, w_3) are each triples of distinct points in Σ then there exists a unique $T \in PGL(2, \mathbb{C})$ satisfying $T(z_j) = w_j$ for $j = 1, 2, 3$.*

Proof. If T_1, T_2 are two elements of $PGL(2, \mathbb{C})$ satisfying $T_1(z_j) = T_2(w_j) = 0, 1,$ and ∞ for $j = 1, 2, 3$ then $T = T_2^{-1}T_1$ sends z_j to w_j for $j = 1, 2, 3$. This element exists since $PGL(2, \mathbb{C})$ is a group. It is unique since if $U \in PGL(2, \mathbb{C})$ satisfies $U(z_j) = w_j$ then T_1 and T_2U send z_j to $0, 1$ and ∞ respectively. Then $T_1 = T_2U$ and $U = T_2^{-1}T_1 = T$. □

Corollary 3.12. *If $T \in PGL(2, \mathbb{C})$ fixes three distinct points of Σ, then T is the identity.*

Proof. If T fixes (z_1, z_2, z_3) then both T and I fix the same set of distinct points and hence $T = I$ by Corollary 3.11. □

Corollary 3.11 shows that $PGL(2, \mathbb{C})$ acts 3-transitively on Σ. We will now examine those $T \in PGL(2, \mathbb{C})$ which leave invariant some subset Δ of Σ containing three elements. These transformations form a subgroup we will call $G(\Delta)$

$$G(\Delta) = \{T \in PGL(2, \mathbb{C}) | T(\Delta) = \Delta\} \leq PGL(2, \mathbb{C}).$$

Now if $\Theta = \{0, 1, \infty\}$ then by the previous results we have that if U is chosen such that $U(\Delta) = \Theta$ we have that $G(\Delta) = U^{-1}G(\Theta)U$. Therefore the subgroup $G(\Delta)$ is isomorphic to the subgroup $G(\Theta)$ and it is sufficient to study the elements which leave Θ invariant.

It follows from Corollary 3.11 that for each permutation π of $\{0, 1, \infty\}$ there exists a unique $T \in PGL(2, \mathbb{C})$, which we call T_π. Thus $\Phi : \pi \to T_\pi$ is an isomorphism between $G(\Theta)$ and the permutation group S_3, i.e. $\theta \cong S_3$. Recall that the permutation group of n elements is denoted by S_n and has $n!$ elements. Thus we need $3! = 3 * 2 * 1 = 6$ elements to form Θ. It is easy to check that these elements are the set

$$G(\Theta) = \left\{z, 1 - z, \frac{1}{z}, \frac{z}{z-1}, \frac{1}{1-z}, \frac{z-1}{z}\right\},$$

which correspond to the elements of S_3

$$S_3 = \{(0)(1)(\infty), (01)(\infty), (0\infty)(1), (0)(1\infty), (01\infty).(0\infty1)\}$$

We now introduce the cross-ratio.

Definition 3.13. If (z_0, z_1, z_2, z_3) is a 4-tuple of distinct elements in Σ the cross ratio

$$\lambda = (z_0, z_1; z_2, z_3)$$

is defined to be $T(z_0)$ where T is the unique element of $PGL(2, \mathbb{C})$ such that $T(z_1) = 0, T(z_1) = 1, T(z_3) = \infty$.

The first thing to note about the cross ratio is that since $z_0 \notin \{z_1, z_2, z_3\}$ then $\lambda \notin \{0, 1, \infty\}$. In fact, we have a formula for $\lambda = T(z_0)$ given by

$$T(z_0) = \frac{(z_0 - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z_0)}$$

with the usual convention of taking limits if some $z_j = \infty$. We now prove a theorem showing when two cross ratios are identical.

Theorem 3.14. *Let (z_0, z_1, z_2, z_3) and (w_0, w_1, w_2, w_3) each be 4-tuples of distinct elements in Σ. Then there exists some $T \in PGL(2, \mathbb{C})$ with $T(z_j) = w_j$ if and only if $(z_0, z_1; z_2, z_3) = (w_0, w_1; w_2, w_3)$.*

Proof. First assume that $T(z_j) = w_j$ for $j = 0, 1, 2, 3$. Let U be the unique element of $PGL(2, \mathbb{C})$ sending w_1, w_2, w_3 to $0, 1, \infty$. Then $U(w_0) = (w_0, w_1; w_2, w_3)$ and UT is the unique element sending z_1, z_2, z_3 to $0, 1, \infty$. Thus $(z_0, z_1; z_2, z_3) = UT(z_0) = U(w_0) = (w_0, w_1; w_2, w_3)$.

Conversely, assume that $(z_0, z_1; z_2, z_3) = (w_0, w_1; w_2, w_3) = \lambda$. Then there exists unique U, V such that $U(z_j) = V(w_j) = \lambda, 0, 1, \infty$ for $j = 0, 1, 2, 3$. Then $V^{-1}U(z_j) = w_j$ for $j = 0, 1, 2, 3$ so take $T = V^{-1}U$. □

So we see that group elements of $PGL(2, \mathbb{C})$ mapping two 4-tuples to each other only exist in the case that their cross ratios are the same. What happens when two 4-tuples are permutations of each other? In general this does not preserve the cross ratio. Let us examine this in detail.

First notice that if z_0, z_1, z_2, z_3 are four distinct elements of Σ we expect that the cross ratio is not preserved under permutation. This is because a permutation will generally not fix z_0 and thus $T(z_0)$. But given some λ by Theorem 3.14 can always find some T_λ such that $T_\lambda(z_j) = \lambda, 0, 1, \infty$. Without loss of generality we can study permutations of the set $\{\lambda, 0, 1, \infty\}$. Since this is a four element set we have that the symmetric group of these elements is isomorphic to S_4. Denote this group G and for each $\pi \in G$ a permutation we define

$$\lambda_\pi = (\pi(\lambda), \pi(0); \pi(1), \pi(\infty)).$$

For example, if $\pi = e$ is the identity permutation then $\lambda_e = (\lambda, 1; 0, \infty)$. On the other hand if $\pi = (\lambda 01\infty)$ then

$$\begin{aligned}
\lambda_\pi &= (0, 1; \infty, \lambda) \\
&= \lim_{z\to\infty} \frac{(0-1)(z-\lambda)}{(1-z)(\lambda-0)} \\
&= \lim_{z\to\infty} \frac{\lambda - z}{\lambda - \lambda z} \\
&= \lim_{z\to\infty} \frac{\lambda/z - 1}{\lambda/z - \lambda} \\
&= \frac{1}{\lambda}
\end{aligned}$$

One might expect to obtain $|G| = 4! = 24$ different values of λ_π for the 24 permutations in G but this is not the case since if we have $\pi' = (\lambda)(0\infty)(1)$ then

$$\begin{aligned}
\lambda_{\pi'} &= (\lambda, \infty; 1, 0) \\
&= \lim_{z\to\infty} \frac{(\lambda-z)(1-0)}{(z-1)(0-\lambda)} \\
&= \lim_{z\to\infty} \frac{\lambda - z}{\lambda - \lambda z} \\
&= \frac{1}{\lambda}.
\end{aligned}$$

And so we have found that $\lambda_\pi = \lambda_{\pi'}$ even though $\pi \neq \pi'$. Therefore the induced homomorphism ϕ which maps π to λ_π is not injective.

We can find that the permutations $\nu_1 = (01)(\lambda\infty), \nu_2 = (\lambda 0)(1\infty), \nu_3 = (\lambda 1)(0\infty)$ and $\nu_4 = e = (\lambda)(0)(1)(\infty)$ form a subgroup $N \leq G$, and for any $\pi \in G$ and $\nu \in N$ we have $\lambda_{\nu\pi} = \lambda_\pi$. For example if $\nu = (\lambda 1)(0\infty)$ and $\pi \in G$ then $\lambda_\nu = (1, \infty; \lambda, 0)$ and

$$\begin{aligned}
\lambda_{\pi\nu} &= \frac{(\pi(1) - \pi(\infty))(\pi(\lambda) - \pi(0))}{(\pi(\infty) - \pi(\lambda))(\pi(0) - \pi(1))} \\
&= \frac{(\pi(0) - \pi(\lambda))(\pi(1) - \pi(\infty))}{(\pi(0) - \pi(1))(\pi(\infty) - \pi(\lambda))} \\
&= \lambda_\pi.
\end{aligned}$$

Thus for each $\pi \in G$ and $\nu \in N$ we have $\pi\nu\pi^{-1} \in N$ and therefore N is a normal subgroup of G. By Lagrange's theorem we find

$$\begin{aligned}
|G : N| &= \frac{24}{4} \\
&= 6
\end{aligned}$$

and so we know there to be 6 different cosets πN in G, each represented by some λ_π. In fact, as we have seen, the permutations $\pi \in G$ which fix λ and permute $\{0, 1, \infty\}$ form a subgroup $G(\Theta) \cong S_3$. We found that this group has 6 elements

$$G(\Theta) = \left\{z, 1-z, \frac{1}{z}, \frac{z}{z-1}, \frac{1}{1-z}, \frac{z-1}{z}\right\}.$$

Since we have $z_0 = \lambda$ we have $\lambda = T(\lambda)$ and

$$H = \left\{\lambda, 1-\lambda, \frac{1}{\lambda}, \frac{\lambda}{\lambda-1}, \frac{1}{1-\lambda}, \frac{\lambda-1}{\lambda}\right\}$$

are 6 values that we can use to represent each coset of S_4/N. We can find that

$$S_4/N \cong H \cong S_3.$$

For each S_n we always have a proper normal subgroup A_n such that $S_n/A_n \cong C_2$ yet in the case of $n = 4$ we find that S_4 has as a proper normal subgroup N. This is the only n for which S_n has a proper normal subgroup other than A_n ([2], Lemma 2.4)!

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